

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

MI-5003-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Mineral Processing

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any Five questions.
ii) All questions carry equal marks.

1. a) Describe the application of thickeners in water reclamation.
b) Explain the operation of a flotation column.
2. a) Explain the theory of ball mill operation.
b) What are the different methods of sampling which are used in Mineral processing plant? What do you understand by Coning and Quartering?
3. a) Write short notes on the following:
Fixed screen plunger jigs (harz jig), vibrating screens, screening efficiency.
b) What is Heavy Media Separation? Explain with schematic sketch the heavy media separation process.
4. a) Discuss the principles and applications of electrostatic separation.
b) Explain washing of coal. How it will be carried out?

MI-5003-CBGS

PTO

[2]

5. a) Define the terms :
 - i) Enrichment ratio
 - ii) Percentage recovery
 - iii) Ratio of concentration
 - iv) Efficiency of screenb) Describe the working principle of a Baum Jig.
6. a) Explain the principle of functioning of a Wilfley shaking table with suitable diagram.
b) Describe the principle of a pneumatic cyclone in separation of ore particles in mineral processing.
7. Describe different types of screen used in mineral industry, their application and limitations.
8. Write notes on the following:
 - a) Dewatering and drying
 - b) Electrostatic separator

MI-5003-CBGS